

Westward Expansion of Melanistic Fox Squirrels (*Sciurus niger*) in Omaha, Nebraska

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Notes and Discussion

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ABSTRACT.—Historically, melanistic fox squirrels have been found in Council Bluffs, Iowa, and along the Missouri River in Eastern Omaha, Nebraska. However, recent anecdotal observations suggest that the melanistic trait in fox squirrels is expanding westward into Omaha. Squirrels were surveyed in the autumn of 2010 and 2011 along transect lines within five major city sections and different habitat types; comparisons were made to a survey performed in 1973. Proportion of melanistic squirrels in Council Bluffs remained at levels similar to 1973 surveys (~50% melanistic). Since 1973 melanistic fox squirrels have increased in Omaha, with a higher proportion of melanistic squirrels in northwest (7.4%) and northeast (7.6%) compared to southwest or southeast Omaha (4.6%). Melanistic squirrels were found in higher proportions in parks (12.1%) and residential (10.9%) habitats compared to business (6.1%), industrial (7.4%), or golf course (8.0%) habitats. Melanistic squirrels were also observed more frequently at colder temperatures than rufus squirrels. The results yielded significant variation in the percent of melanistic individuals in each section and suggest the proportion of melanistic individuals is increasing westward.

INTRODUCTION

Numerous animals show polymorphic melanism in their coloration, including snakes (Mooi *et al.*, 2011), numerous lizards (Engelbrecht *et al.*, 2011; Portik *et al.*, 2010; Weijola and Sweet 2010), leopards (*Panthera pardus*; Kawanishi *et al.*, 2010), tundra voles (*Microtus oeconomus*; Jung *et al.*, 2009), white-tailed deer (*Odocoileus virginianus*; Baccus and Posey, 1999), woodrats (*Neotoma* spp; Stangl *et al.*, 1999), and eastern fox squirrels (*Sciurus niger*; Kiltie 1992a, b). Melanism occurs from changes in a single locus, dominant allele that is responsible for the black color coat observed in melanistic animals (van Hof *et al.*, 2011). Melanism is likely associated with mutations in the melanocortin 1 receptor (*MC1R*) gene which is associated with coat color changes in many vertebrates, such as, common mice, rock pocket mice, horses, cattle, birds, and canids (Dreger and Schmutz, 2010; McRobie *et al.*, 2009; Anderson *et al.* 2009). Similarly, melanism within tree squirrels has been found to be associated with a base pair sequence deletion within the *MC1R* gene (McRobie *et al.*, 2009) resulting in incomplete dominance for coat color. The phenotypic expression of melanism involves selection for various coat colors and is a useful characteristic for studying evolutionary changes (Majerus and Mundy, 2003). For example melanism may provide a selective advantage against predator-prey encounters in dark environments (Majerus and Mundy, 2003) or thermal advantages as a result of increased absorption of solar radiation (Fratto and Davis, 2011).

A large tree-dwelling squirrel in North America, the eastern fox squirrel (*Sciurus niger*), is found throughout the eastern United States and includes, at the western edge of its range, Omaha, Nebraska and Council Bluffs, Iowa. The normal color phase for this species is rufus or reddish orange; however, some individuals' exhibit a melanistic phase in which they are completely black. Historically, individuals exhibiting the melanistic phase have been found in Council Bluffs, Iowa and into extreme eastern Omaha (Lueninghoener, 1973). However, recent anecdotal observations suggest the proportion of melanistic individuals is increasing westward into other parts of Omaha. Because the proportion of melanistic squirrels is expanding, there may be a selective advantage that contributes to increased fitness over rufus squirrels.

Within fox squirrels, the melanistic pattern differs throughout their range with eastern populations having black heads with white noses (Kiltie, 1992a) and populations in the Mississippi River basin lacking the white patches but having body coloration ranging from agouti to solid black (Kiltie, 1989; Kiltie, 1992b). Melanistic fox squirrels in the metropolitan area of Omaha are typically solid black; however, some atypical forms have been reported, including black squirrels with orange tails and squirrels with an orange and black bodies (Lueninghoener, 1973).

We surveyed the Omaha metropolitan area for the proportion of melanistic and rufus fox squirrels and compared our results to a past survey (Lueninghoener, 1973) to determine whether the proportion of melanistic fox squirrels has increased. In addition we measured the proportion of melanistic squirrels in the various city habitats found throughout Omaha. Several studies have looked at melanistic squirrels

in the context of predator avoidance or cryptic coloration (Kiltie, 1989; Kiltie, 1992a, b); however, in this study we look at the potential thermal benefits of melanism. We also revisit a survey (Lueninghoener, 1973), conducted in 1973, of melanistic squirrels in the Omaha metropolitan area to identify temporal changes in the proportion of melanistic squirrels; making this study one of the first to measure changes in the proportion of melanism in a population through time.

METHODS

Study area.—Visual surveys for fox squirrels were conducted in the Omaha, Nebraska and Council Bluffs, Iowa metropolitan area (41.2586°N, 95.9375°W). Omaha and Council Bluffs consist of a contiguous city surrounded by farmland. The Missouri River is a natural barrier dividing Omaha from Council Bluffs; however, numerous vehicle, pedestrian, and train bridges span the river, effectively connecting the two cities. Omaha consists primarily of residential housing with numerous retail and business locations, parks, golf courses, cemeteries, and industrial and railway sectors along the river.

Squirrel surveys.—Omaha and Council Bluffs were divided into five major city sections: Council Bluffs, Northwest, Northeast, Southwest, and Southeast Omaha. Council Bluffs consisted of the western section of the city, west of state highway 192. Omaha was divided into four quadrants by using 72nd Street and Dodge Street as cardinal dividing lines. Each of the five major city sections was further divided into grids of ~1.6 km²; within each survey grid, 4–5 streets were marked as survey routes for squirrels. Survey routes were of similar length (~0.5 km) between all city grids; however, some variation in survey route length occurred due to the unique aspects of city planning.

Habitat types encountered during surveys included: residential, business, park, golf course, cemetery, and school. Residential habitat included single home dwellings and mobile homes with surrounding yards. Golf courses, parks, and cemeteries are all typically dominated by large expanses of cultivated and mowed grass; however, these habitats differed by secondary vegetation and patterns of use by humans. Golf courses consisted of long narrow stretches of grass lined with tall trees with seasonal (warm weather) low density human use. Cemetery were open grassland with most trees lining the periphery of the cemetery, or along major roads and low, sporadic human use. Schools had large grass expanses frequently used as sporting fields (soccer, football), with trees interspersed with school buildings and high human use. Business consisted of strip malls and parking lots, whereas industrial were large warehouses and heavy equipment lots.

Surveys were conducted using a modified version of the spot count method established by Lueninghoener (1973) and Hicks (1942) and were conducted in the Fall (Sep. and Oct.) 2010–2011 during the morning or afternoon, to avoid mid-day when squirrels tend to be inactive (Koprowski, 1994). Fox squirrels show a bimodal activity pattern with high activity from just after sunrise to mid-day and a second active period from early afternoon to just before sunset (Koprowski, 1994), making morning and afternoon counts equally valid. Surveyors walked each transect at a normal pace on both sides of the street watching for squirrels. When a squirrel was observed the color was noted and designated as being rufus (orange) or a melanistic (black) and was added to the count for that street. In addition the habitat type and weather were noted. Only live squirrels were counted, but dead squirrels were also noted.

Distribution of melanistic squirrels was analyzed using a Negative Binomial Regression of the percent melanism with city section and habitat as treatment factors. This regression analysis is useful in analyzing data that are nonparametric in nature, such as survey data. PROC GENMOD was used to calculate Chi-square values for comparisons among the main factors (Year, Section, and Habitat; SAS 2002). If there were no significant interactions, differences in the main factors were compared using the PDIF option in the LSMEANS statement. Differences in terms with significant interactions were compared using the SLICE option in LSMEANS. Unpublished raw data relating melanistic and rufus squirrel observations to ambient temperature was listed in the appendix of Lueninghoener (1973) and was used, in this study, to generate a regression analysis for ambient temperature vs. percent melanistic activity. All data are presented as mean $\pm$ standard error, and all differences were considered significant at the $P = 0.05$ level.

RESULTS

Fox squirrels were most easily observed in Council Bluffs, followed by the southern (SE and SW) sections of Omaha (Fig. 1). The number of melanistic individuals observed during surveys was greater

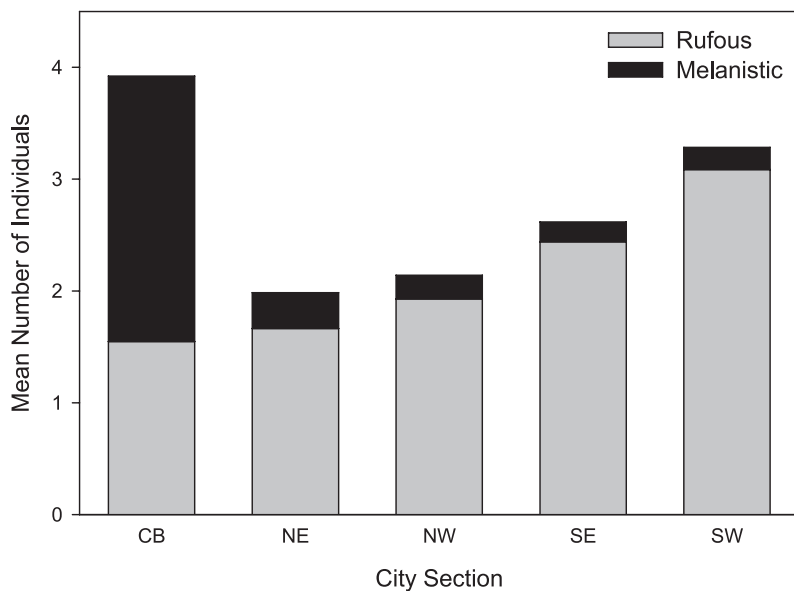


FIG. 1.—Average number of rufus and melanistic fox squirrels observed during survey counts across the five major city sections of the Omaha and Council Bluffs metropolitan area

in Council Bluffs compared to all other sections of Omaha. Distribution of melanistic squirrels in the Omaha/Council Bluffs metropolitan area differed significantly by city section ($\chi^2_4 = 210.65$, $P < 0.0001$; Fig. 2). The city section which showed a significantly higher concentration of melanistic squirrels was Council Bluffs, Iowa with 50.8% melanistic squirrels. Within Omaha, the northeast (7.6%) and northwest (7.4%) had relatively higher proportions of black squirrels compared to southeast and southwest Omaha. Southeast (4.6%) and southwest (4.6%) Omaha had significantly fewer black squirrels than other sections of the Omaha/Council Bluffs metropolitan area.

A difference in the distribution of melanistic squirrels was found among habitats ($\chi^2_6 = 23.93$, $P = 0.0005$). Black squirrels were found in significantly higher proportions in parks (12.1%) and residential (10.9%) areas compared to business (6.1%), golf courses (8.0%), and industrial areas (7.4%; Fig. 3).

There was a significant interaction between city section and habitat as well ($\chi^2_{19} = 51.84$, $P < 0.0001$). Melanistic fox squirrels were found in high proportions in almost all habitat types within Council Bluffs, Iowa; however, melanistic squirrels were typically found in parks, golf courses, residential, and school habitats in Omaha. Within the northeast Omaha city section parks had a significantly higher proportion of melanistic squirrels (26.1%) compared to other habitats within the northeast Omaha city section. There were proportionately fewer melanistic squirrels in the school habitat of the northwest (1.2%), northeast (2.5%), and southwest (0.5%) Omaha city sections. Although the proportion of melanistic squirrels was high (>30%) in habitats within Council Bluffs, the variability was also consistently high between habitats. The percent of active squirrels that were melanistic increased significantly ($F_{1,10} = 8.17$, $P = 0.02$) with decreasing ambient temperatures (Slope = -1.02 , $R^2 = 0.45$, $P = 0.02$; Fig. 4).

DISCUSSION

Over the last 40 y, the proportion of melanistic squirrels has been expanding in the Omaha and Council Bluffs metropolitan area (Table 1). Results of the Lueninghoener (1973) survey showed melanistic squirrels were only found in Council Bluffs, Iowa and along the Missouri River of Omaha, corresponding to the northeast and southeast Omaha sections of this study. Within Council Bluffs, Lueninghoener (1973) found the proportion of melanistic squirrels was 45.2%; whereas, this study

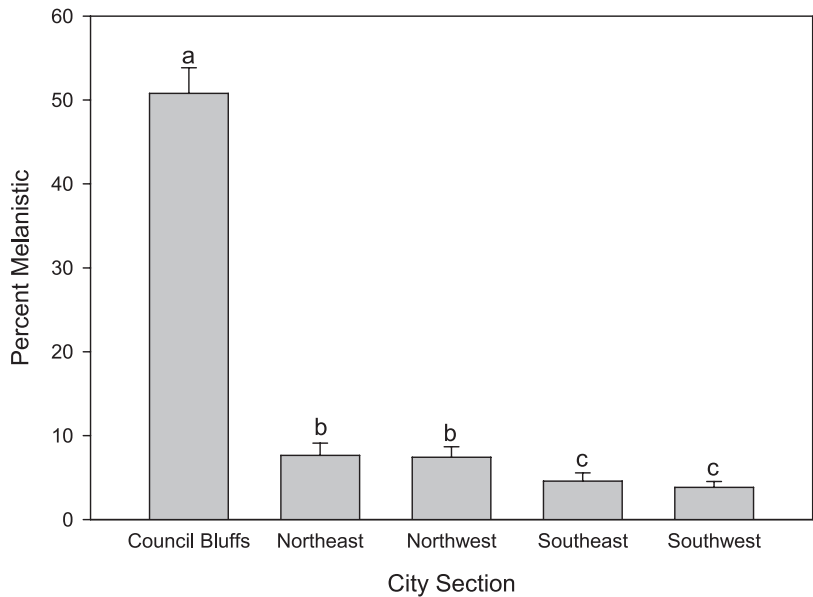


FIG. 2.—Distribution (percent melanistic) of black fox squirrels across the five major city sections of the Omaha and Council Bluffs metropolitan area. Significant differences are indicated with different letters

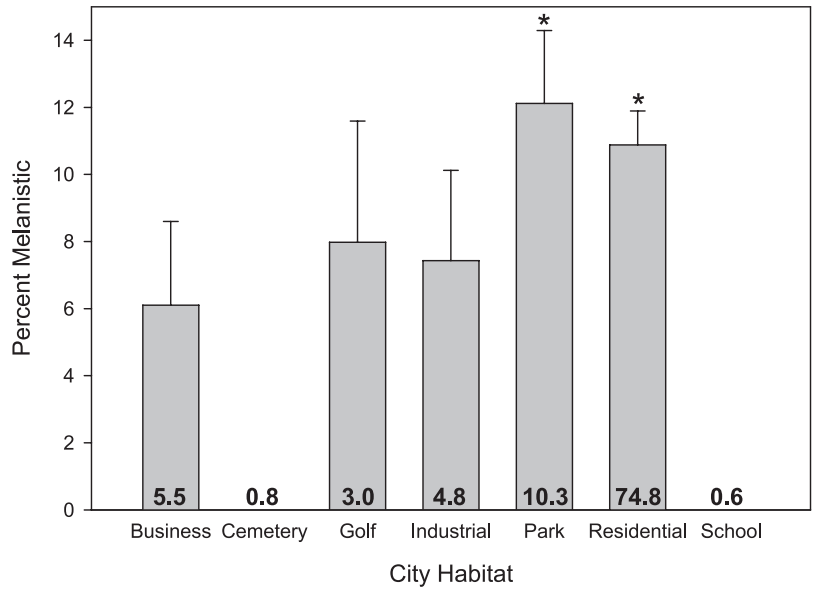


FIG. 3.—Distribution (percent melanistic) of melanistic fox squirrels across habitat types of the Omaha and Council Bluffs metropolitan area. Numbers at the bottoms of the bars indicate the percent of city survey transects that were composed of that habitat. Significant differences are indicated with asterisks

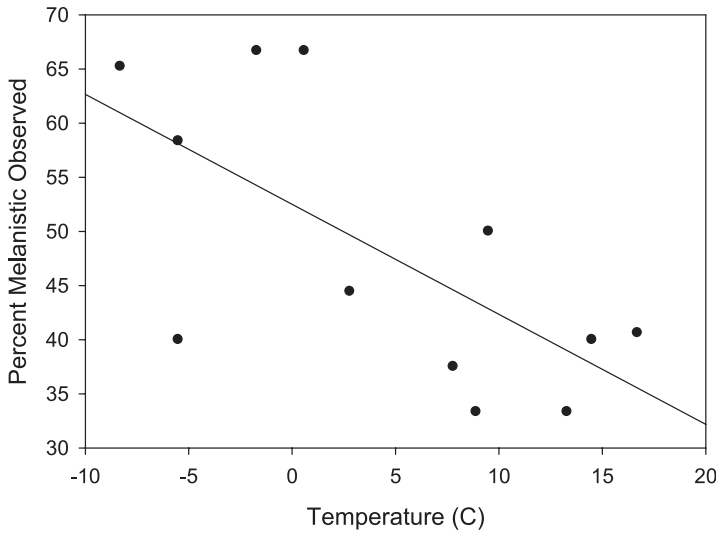


FIG. 4.—Proportion of melanistic fox squirrels active at different ambient temperatures. Raw data were obtained from Appendix III of Lueninghoener (1973)

found melanistic squirrels averaged $50.8 \pm 3.0\%$. Although the proportion of melanistic squirrels increased slightly in Council Bluffs when compared to their historic frequency, they have expanded westward and have increased within all sections of Omaha.

The current study found that melanistic fox squirrels have expanded to inhabit the entire Omaha metropolitan area. In particular melanistic squirrels have increased to almost twice the proportion in northern sections of Omaha (7.6%) compared to southern sections (4.6%). Both northwestern and southwestern sections of Omaha now contain melanistic individuals, which were absent in 1973 (Lueninghoener, 1973). In addition to Omaha, melanistic fox squirrels have even been observed in Lincoln, Nebraska, 45 mi west of Omaha (personal observation).

Specific habitats differed in squirrel abundance, with those habitats (residential, school, golf course) containing suitable tree squirrel habitat (big trees) having fox squirrels, and those without suitable habitat (business and industrial) generally lacking fox squirrels. Fox squirrels prefer urban habitats with large canopy trees, such as oak, and avoid open grass areas, except when actively caching food or travelling between trees (McCleery *et al.*, 2006). Similarly, buildings were only selected for use during late winter and early spring, probably as a refuge against the cold (McCleery *et al.*, 2006). Areas that consisted entirely of concrete, buildings, and asphalt did not support fox squirrels (McCleery *et al.*, 2006).

TABLE 1.—Comparison of the percent of melanistic squirrels observed in this study and in a study conducted by Lueninghoener (1973)

City section	Lueninghoener 1973	Current study
Council Bluffs	45.2%	50.8%
NW Omaha	<5.0%	7.4%
NE Omaha	<5.0%	7.6%
SW Omaha	0.0%	4.6%
SE Omaha	0.0%	4.6%

Forty years ago, melanistic fox squirrels were present in rural areas of both Council Bluffs and eastern Nebraska (*i.e.*, southeast and northeast Omaha), but were not present in southwest and northwest Omaha (Lueninghoener, 1973). This was likely a result of the lack of large trees in these areas. In Council Bluffs Lueninghoener (1973) found that the distribution of fox squirrels was also related to large trees, with locations found in the relatively treeless, floodplain having fewer squirrels (0.20 squirrels/block) compared to city blocks found within the Loess Hills (1.01 squirrels/block), a unique upland riparian habitat characterized by forests. Large trees were also found to be the major determining factor in the presence of fox squirrels in urban environments in Texas (McCleery *et al.*, 2006). Although we did not measure changes in the extent of the Omaha metropolitan area, or changes in the number of large trees within the city, Omaha has experienced considerable westward growth and has recently incorporated an outlying city (Elkhorn, NE) into the Omaha city limits. It is likely that the westward expansion of the city of Omaha, which includes a concurrent expansion of large trees, has enabled tree squirrels, including melanistic individuals, to expand westward.

Melanistic variants are clearly increasing in numbers throughout the Omaha and Council Bluffs metropolitan area. In addition melanistic fox squirrels have been observed in several locations in Lincoln, Nebraska. There have been several relocation/importation attempts that have taken melanistic squirrels from the Council Bluffs area and released them in cities in central Nebraska (Fremont, Fairbury, and Hastings, NE, and Marysville, KS; Lueninghoener, 1973). Relocation attempts in Nebraska are the result of citizens moving melanistic squirrels on their own volition and are not part of any official wildlife policy. These "citizen relocations" suggest there is some human affinity for the melanistic squirrels. However, there must be some advantage for melanism to increase in the population following introduction.

Several suggestions have been made to account for the advantage of melanism, including predator avoidance (Kiltie, 1992a, 1989), cryptic coloration (Kiltie, 1992b), and thermoregulation (Armitage, 2009). Kiltie (1992a) suggested melanism in fox squirrels was an adaptation for cryptic coloration, as a response to blackened habitat associated with frequent forest fires in the longleaf pine forests of the southeastern U.S. Similarly, Kiltie (1992b) also looked at cryptic coloration in Mississippi populations. Although some evidence for a cryptic advantage was found by Kiltie (1992b), it was not overwhelming and left the possibility of other factors that may lead to an increase in melanism in fox squirrels. It is unlikely that melanistic individuals inhabiting the Omaha metropolitan area are experiencing benefits from cryptic coloration as there is virtually no fire to blacken the habitat and the squirrels mostly move about on green manicured lawns or pavement.

Melanism has been implicated as a thermal benefit to animals, especially ectotherms, with increasing melanism associated with cooler temperate latitudes (Clusella *et al.*, 2007; Alho *et al.*, 2010). There is evidence that melanistic snakes warm faster than normal colored snakes (Tanaka, 2005) which provides a selective advantage to melanistic individuals allowing them to become active earlier and remain active longer. There is even evidence of daily and seasonal color changes in boa constrictors (Boback and Siefferman, 2010) which may be under physiologic control by the individual. In other snakes warming rates are unaffected, but equilibrium temperature may be higher in melanistic individuals (Bittner *et al.*, 2002). However, the thermal benefit of melanism in snakes does not seem to provide enough selective pressure to increase body size (Tanaka, 2009a) or alter preferred body temperatures (Tanaka, 2009b).

Within endothermic species the thermal advantage of melanism may be less important, but some thermal benefits of melanism have been ascribed to endotherms as well (Sirkiä *et al.*, 2010). Armitage (2009) describes the thermal ecology of marmots (*Marmota* spp.), which can have high proportion of melanism, ranging from 6–29% melanistic, stating that lighter colored marmot species occur in warmer habitats and that marmots have been observed orienting their bodies toward the sun and basking during colder portions of the day (morning and evening) while becoming inactive in their burrows during the heat of midday. Similarly, in the pied flycatcher (*Ficedula hypoleuca*) there was increased reproductive output from melanistic individuals during periods of cold temperatures (Sirkiä *et al.*, 2010). Darker color morphs sang more frequently in colder weather (Ilyina and Ivankina, 2001); females paired to darker males had larger clutches (Sirkiä *et al.*, 2010); and dark males were more active in colder weather, potentially providing more food to nestlings than lighter colored males (Kerimov *et al.*, 2006).

Winter is the most critical time for fox squirrels, with energy resources being limited to stored energy in the form of fat and cached food (Husband, 1976). Although individuals are typically able to store enough energy to get them through winter, competition for food resources, extreme winter conditions, and mast crop failure may generate times of energy stress. During these times, melanistic individuals, who may spend less energy warming or may gain additional foraging time from their black coloration, may gain a considerable advantage.

Fox squirrels have activity patterns that are unimodal in winter centering around the warmest part of the day and bimodal in summer with activity highest during the coolest part of the day (morning and evening; Koprowski and Corse, 2005; Jodice and Humphrey, 1992; Nelson, 1981; Brown and Yeager, 1945). Accordingly, melanistic squirrels should increase their times of activity during colder periods, giving them longer active periods in winter. Lueninghoener (1973) suggested that a higher proportion of melanistic individuals were active on days that were colder than 7°C and, in fact, data he collected supports this claim (Fig. 4).

Although cryptic coloration and thermal advantage are the two most common hypotheses for the selective advantage of melanism other, as of yet untested, factors may confer substantial benefits to melanistic fox squirrels. Additional hypotheses for the advantage of melanism might also include behavior (melanistic squirrels might be socially dominant), roadkill avoidance (melanistic squirrels are easier to see while driving), or an increased “cute factor” (melanistic squirrels being deemed more valuable by the public and hence are fed and cared for more than rufus individuals). From our study it is clear that melanistic fox squirrels are increasing throughout the Omaha metropolitan area, but more study is needed to tease apart which of these hypothesized advantages are indeed true.

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